Marketing Mix Modelling & Budget Optimisation

Executive summary

This project evaluates advertising effectiveness across TV, Facebook/Instagram, Google Search, YouTube, Radio and Outdoor, and uses the resulting Marketing Mix Model (MMM) to propose a more efficient allocation of a fixed advertising budget. The analysis uses 36 monthly observations from a supplied synthetic marketing dataset. Observed sales total LKR 3,063.31 m and advertising expenditure totals LKR 668.74 m over the study period.

Data preparation retained every month, corrected small rounding differences in total advertising expenditure, documented outliers and created an analysis-ready dataset. Exploratory analysis identified strong seasonal and promotional associations with revenue, as well as substantial collinearity among price, inflation and time indicators.

The selected model, M3a, combines non-negative media coefficients, ridge regularisation, geometric adstock, Hill saturation and five external controls. It achieved rolling-origin cross-validation RMSE of LKR 2.956 m and held-out test RMSE of LKR 1.750 m, MAE of LKR 1.563 m and R^2 of 0.887. The last six months were held out from model fitting and hyperparameter selection. After evaluation, the selected specification was refitted on all 36 months to generate the attribution and optimisation outputs.

Outdoor, YouTube and Google Search have the highest estimated historical channel ROAS: 6.12x, 4.17x and 4.08x respectively. Search and YouTube rank strongly across alternative models; Outdoor's exact ranking is more sensitive to specification. Radio's zero fitted contribution is an uncertain model estimate and does not establish zero real-world effectiveness.

With the 2025 average monthly budget fixed at LKR 20.024 m, constrained optimisation increases estimated steady-state incremental monthly revenue from LKR 48.027 m to LKR 50.143 m: an increase of LKR 2.116 m, or 4.41%. Portfolio modelled ROAS rises from 2.398x to 2.504x. This is a model scenario, not an observed sales improvement or guaranteed forecast. The recommended action is a controlled pilot of gradual reallocations towards Outdoor, Search and YouTube, subject to confirmed business constraints.

The delivered dashboard is a Tableau packaged workbook with supporting screenshots. This differs from the original Power BI requirement and requires acceptance of the platform change or equivalent delivery in Power BI. Revenue-based ROAS is reported throughout because reliable profit margins are unavailable.

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1. Introduction and objectives

Marketing expenditure must support sales while recognising that channels differ in reach, timing and efficiency. Simple comparisons between total revenue and advertising spend cannot isolate media effects from promotions, demand seasonality, competitive activity or economic conditions. Similarly, allocating more budget to the channel with the highest historical average return ignores diminishing marginal returns.

This project therefore develops an integrated workflow linking data engineering, exploratory analysis, MMM, channel effectiveness analysis, constrained optimisation and dashboard reporting. Its objectives are to:

- Prepare a validated dataset and preserve chronological order in evaluation.
- Identify sales patterns, channel spending behaviour and relevant external factors.
- Compare regression models that account for carryover and diminishing returns.
- Estimate channel contributions, average ROAS and marginal ROAS with uncertainty.
- Allocate a fixed budget under channel spending limits and examine alternative scenarios.
- Communicate the findings, assumptions and recommendations through a dashboard and this report.

The work is an observational modelling exercise on synthetic data. Its findings support hypothesis formation and investment testing; they do not independently establish causal advertising effects.

2. Dataset and analytical workflow

The source is `data/raw/Dataset.xlsx`, sheet `MMM_Data`. It contains 36 consecutive monthly records, January 2023 to December 2025, and 25 original columns. The target is nominal monthly `sales_revenue_lkr`. The six media predictors record monthly channel expenditure in LKR.

Variable group	Contents and analytical purpose
Media expenditure	TV, Facebook/Instagram, Google Search, YouTube, Radio and Outdoor
Business outcomes	Sales revenue and units sold; revenue is the modelling target
Funnel measures	Impressions, clicks and conversions, used for diagnostics
Commercial and external factors	Price, promotional discount, competitor pressure, economic sentiment, exchange rate and inflation proxy
Calendar and demand	Month, seasonal demand index and public holiday count
Metadata	Country, brand category, region scope and data type; constant in this dataset

The cleaned dataset contains 36 rows and 22 columns. Feature engineering produces 66 columns: the 22 cleaned columns plus 44 derived features. These are a feature catalogue, not 66 simultaneous regression predictors. A column-level dictionary records definitions, data types, missing values and modelling roles.

The workflow is: source workbook → cleaning and validation → EDA → chronological feature preparation → baseline and enhanced MMM comparison → selected model and response functions → channel analysis and constrained optimisation → dashboard and final report. Python modules and notebooks provide the computational record; saved CSV and JSON outputs supply the numerical evidence used here.

3. Data preprocessing and quality assurance

Dates were converted to month-start values and records organised chronologically. The raw workbook contained no missing values, duplicate rows or duplicate months. All 36 observations were retained.

Quality issue	Finding	Treatment
Missing values	None in raw data	No imputation required
Duplicate records/months	None	No deletion required
Total spend reconciliation	16 monthly totals differed from channel sums by at most LKR 2	Replaced totals with the exact six-channel sum
Constant metadata	Four constant columns	Removed from analytical table
Outliers	Six flags across six months	Retained and documented
Temporal continuity	All 36 expected months present	Preserved for lag and carryover modelling

The reconciliation tolerance treats differences up to LKR 5 as rounding; larger inconsistencies stop the cleaning pipeline. The observed differences fall within that tolerance. The source workbook remains unchanged.

Outlier screening combined the IQR rule and absolute z-score checks. Three revenue observations and the recurring December seasonal-demand peaks were flagged. Revenue flags included January 2023 (LKR 68.7 m), April 2023 (LKR 100.7 m) and August 2024 (LKR 69.9 m). The seasonal index reached 1.24 in each December. These observations are commercially plausible, and deletion would reduce an already small sample and interrupt the time series. They were retained with `n_outlier_flags` and a separate flag file.

The saved quality summary records 16/16 cleaned-data checks and 8/8 engineered-feature checks passing. Checks cover valid dates and ranges, non-negative expenditure, positive revenue, exact total-spend reconciliation, funnel consistency, temporal coverage and separation of training and testing.

4. Exploratory data analysis

4.1 Sales and advertising expenditure

Monthly revenue averages approximately LKR 85.1 m, with standard deviation LKR 7.6 m and a range of LKR 68.7 - 100.7 m. Annual monthly averages are LKR 85.7 m in 2023, LKR 82.8 m in 2024 and LKR 86.8 m in 2025. The estimated linear trend is small and statistically inconclusive: approximately LKR 0.12 m per month, $R^2 = 0.03$ and $p = 0.35$ [3].

December and April have the highest average revenue, at approximately LKR 96.7 m and LKR 95.9 m. These patterns support explicit seasonal controls rather than attributing festive sales increases entirely to advertising.

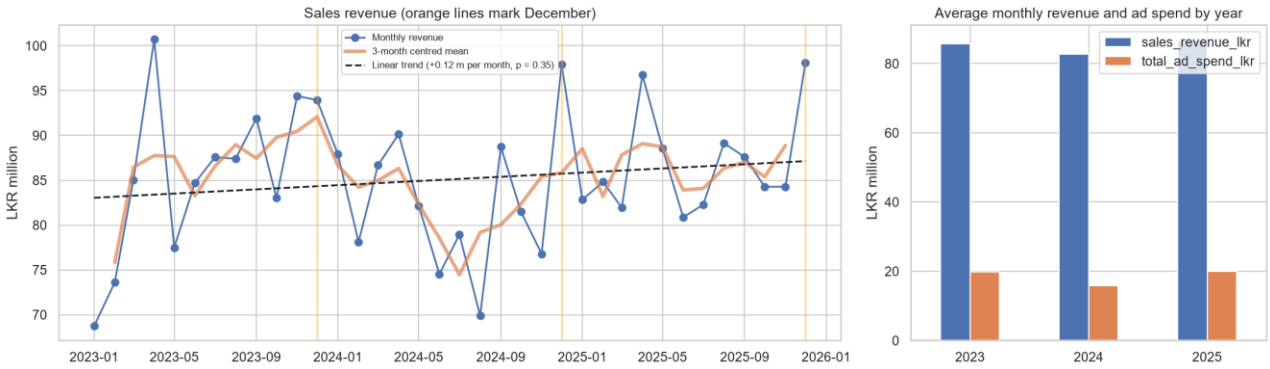


Figure 1. Monthly sales revenue over the study period.

Average monthly advertising expenditure is approximately LKR 18.6 m. Annual spend totals are approximately LKR 238.1 m, LKR 190.3 m and LKR 240.3 m for 2023, 2024 and 2025. Facebook/Instagram and TV receive the largest historical shares, 23.7% and 23.5%; Radio receives 9.0%. Similar channel mixes across years constrain the independent variation available to identify separate effects.

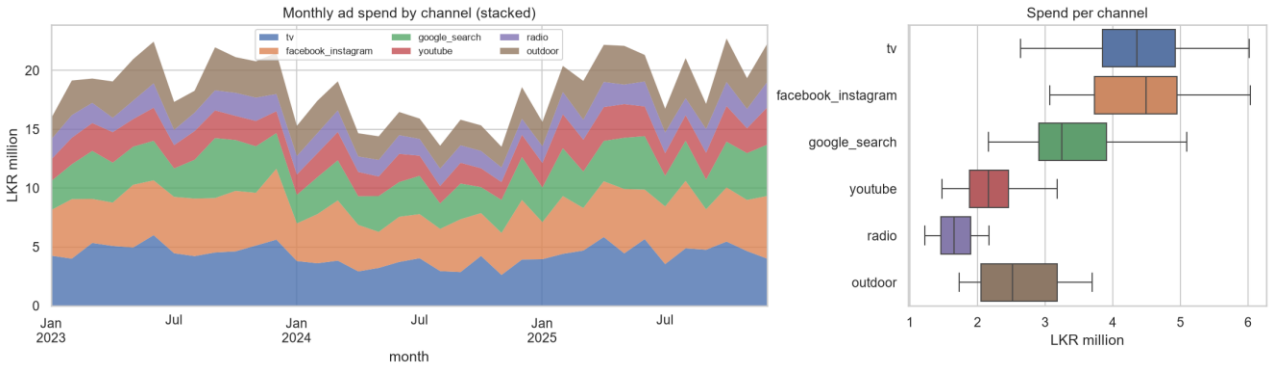


Figure 2. Advertising expenditure by channel.

4.2 Correlations and external factors

Seasonal demand (r = 0.61) and promotional discount (r = 0.58) show the strongest simple positive correlations with revenue. Total advertising spend correlates at 0.45. Individual channel correlations are Outdoor 0.44, Search 0.38, YouTube 0.37, Facebook/Instagram approximately 0.33, TV approximately 0.325 and Radio approximately 0.04. Competitor pressure correlates negatively at approximately −0.335 [3]. These are unadjusted associations, not channel contribution estimates.

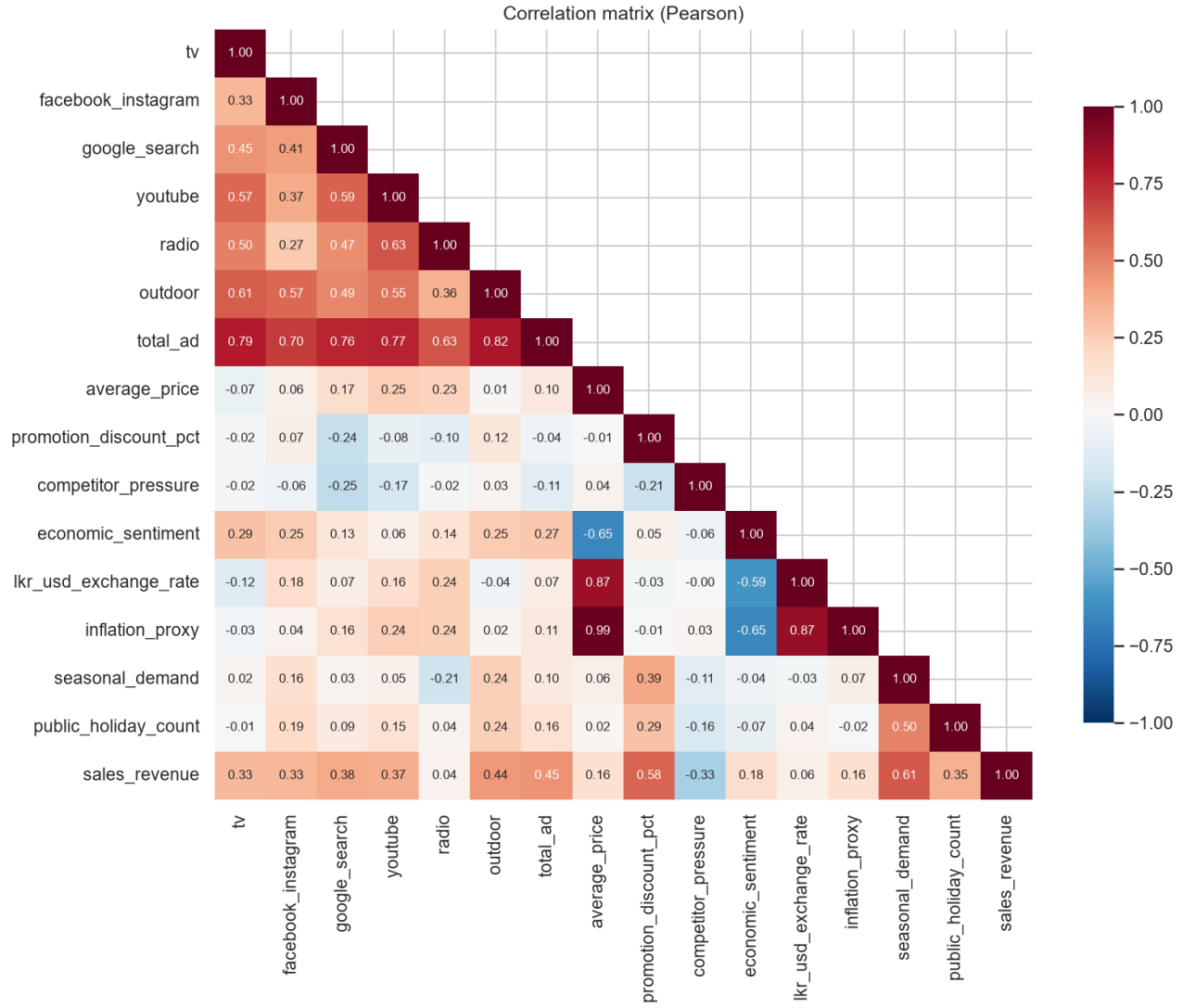


Figure 3. Correlation matrix for media, outcomes and external factors.

Channel-spend correlations range approximately 0.27–0.63, with reported VIFs of 1.6–2.4. Price, inflation and the time index correlate approximately 0.99–1.00, making their simultaneous inclusion unstable. The modelling stage therefore selects a representative economic control.

High-promotion months (discount at least 15%) have 6.8% higher average revenue than other months; months with at least six public holidays have 7.1% higher revenue. Their reported p-values, 0.074 and 0.055, indicate suggestive evidence within this small sample. Calendar and commercial effects must be considered jointly with media.

4.3 Delayed effects and diminishing returns

Most channels show their highest raw revenue correlation in the same month. Radio's raw correlation peaks at a two-month lag, while partial-correlation analysis gives a one-month peak after adjustment for selected controls. These differences motivate estimated carryover parameters; they do not determine an exact causal delay.

Simple logarithmic fits improve over straight-line fits only slightly for Search, YouTube and Outdoor. Channel expenditure varies within relatively narrow ranges, so EDA alone cannot establish the shape of saturation. Adstock and saturation are estimated within the regularised MMM instead.

5. Feature engineering and chronological evaluation

Derived features cover six time variables, 14 spend-shape variables, 12 channel lags, seven efficiency measures and five external/regime measures. Time variables include year, quarter, month, a trend index and sine/cosine seasonal terms. Channel lags use only earlier months. Fixed thresholds define high promotion and high holiday flags.

Lag and rolling features have expected missing values at the beginning of the series. These were left unfilled to avoid using later observations to reconstruct earlier values. No such missingness is present in the raw source measures used by the final model.

Revenue-derived variables, such as blended ROAS and revenue growth, are excluded from revenue predictors. Total advertising spend is excluded when its six component channels are included. Funnel outcomes are reserved for descriptive analysis because conditioning on downstream advertising outcomes can distort attribution.

Evaluation set	Observations	Period
Training	30	January 2023–June 2025
Held-out test	6	July 2025–December 2025

The preprocessing hand-off documents four non-overlapping three-month expanding validation windows, beginning with 18 training months. The MMM subsequently uses a denser rolling-origin scheme inside the same 30-month training set: an 18-month initial window, three-month validation horizon and one-month step, producing ten overlapping folds. These are distinct validation schedules; neither uses the final six-month test period for tuning.

Scaling and transformation parameters are estimated within training operations. The six held-out months assess the selected train-only fit. The later full-history refit supplies operational response functions and historical decomposition; its fitted predictions must not be presented as independent test evidence.

6. Marketing Mix Model methodology

6.1 Baseline and enhanced specifications

The model sequence begins with ordinary least squares (M0), then introduces non-negative media coefficients with ridge regularisation (M1), geometric adstock (M2) and Hill saturation (M3). Additional candidates compare control sets and a flexible saturation exponent. The selected M3a adds the price index to the core controls.

For channel c and month t, normalised geometric adstock can be written as:

$A[c,t] = (1 - lambda[c]) \times x[c,t] + lambda[c] \times A[c,t-1]$.

Here x is expenditure and lambda is the carryover fraction. Initial adstock assumes pre-sample spending was at the first observed month's level. Normalisation preserves spend units at a constant-spending steady state.

Scaled adstock is $s[c,t] = A[c,t] / scale[c]$. The selected Hill response is:

$H[c,t] = s[c,t] / (s[c,t] + K[c])$.

The exponent is fixed at one, producing a concave response with diminishing returns. K controls half-saturation relative to the channel scaling level. The revenue equation is:

`revenue[t] = intercept + sum(beta[c] × H[c,t]) + sum(gamma[k] × control[k,t]) + error[t]`, with `beta[c] ≥ 0`.

Ridge regularisation discourages unstable coefficient estimates. The media non-negativity constraint prevents the fitted model from attributing negative incremental sales to a channel, while still allowing a zero estimate.

6.2 Control selection and hyperparameter search

The final controls are promotion discount, seasonal demand, competitor pressure, economic sentiment and average price index. Inflation and the explicit trend are excluded because of their near collinearity with price. Holiday count is excluded because it worsened validation performance after seasonal demand was included. These decisions reflect validation results and identifiability constraints, rather than a claim that excluded variables are economically irrelevant.

Channel-type bounds constrain carryover: TV 0.3–0.8, Outdoor 0.2–0.7, YouTube 0.1–0.6, Radio 0.1–0.5, Facebook/Instagram 0–0.4 and Search 0–0.3. SciPy Differential Evolution selects carryover, saturation and ridge settings against rolling-origin CV RMSE. The reported selection rule requires the lowest CV RMSE and at least a 2% improvement over standard M3. M3a satisfies that rule. The six-month holdout was reserved for evaluation.

7. Model evaluation and reliability

RMSE measures the typical scale of prediction error with extra weight on large errors. MAE measures average absolute error. R^2 compares prediction error with a mean-outcome benchmark; MAPE expresses absolute error relative to observed revenue. All monetary errors below are LKR millions, and test metrics refer to July–December 2025.

Model	CV RMSE	Test RMSE	Test MAE	Test MAPE	Test R^2
M0 OLS baseline	6.379	3.480	2.972	3.39%	0.554
M1 Constrained linear ridge	3.209	2.505	2.425	2.75%	0.769
M2 Adstock	3.092	1.980	1.853	2.09%	0.856
M3 Adstock + saturation	3.077	2.043	1.901	2.14%	0.846
M3a + price index: selected	**2.956**	**1.750**	**1.563**	**1.77%**	**0.887**
M3b All six controls	2.965	1.993	1.895	2.15%	0.854
M3c Tuned Hill exponent	3.014	2.131	1.966	2.21%	0.833
Alternative A: log-spend ridge	3.408	2.152	1.820	2.06%	0.830
Alternative B: lagged-spend ridge	3.855	1.155	1.004	1.13%	0.951

M3a reduces test RMSE by approximately 49.7% relative to the OLS baseline. Alternative B has the best holdout score but materially worse CV performance than M3a. Selecting it because of that one

six-month result would undermine the intended separation between selection and testing. Its negative Radio coefficient also complicates attribution. M3a remains the downstream model because of its CV performance and interpretable response functions.

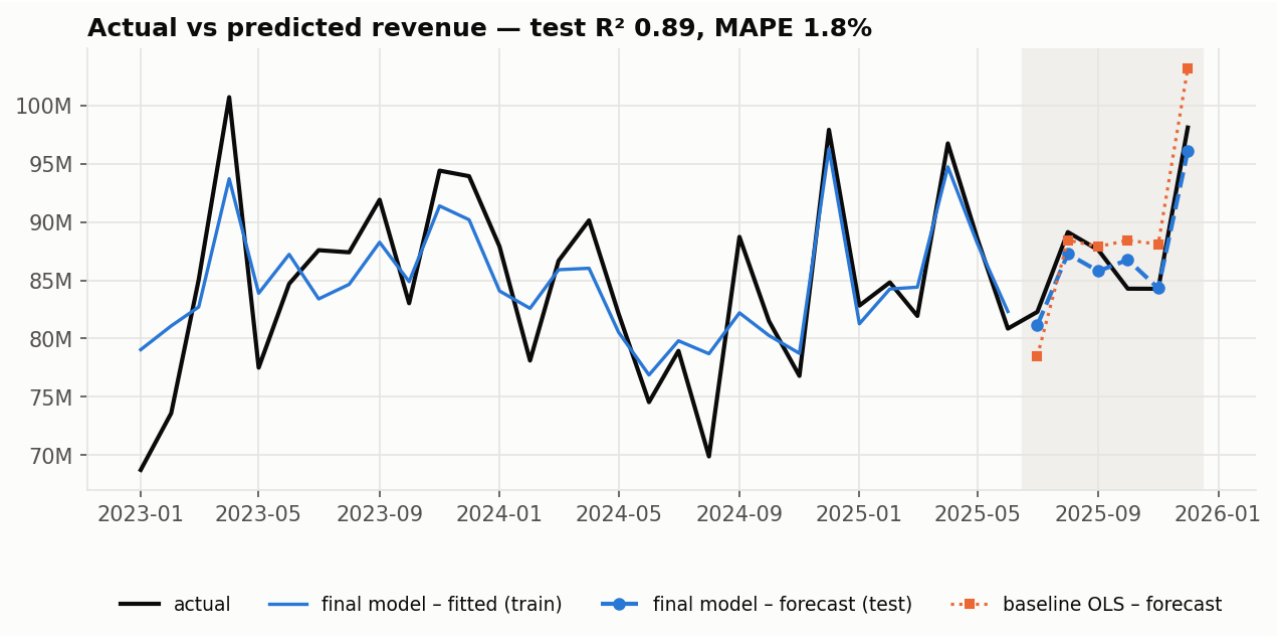


Figure 4. Actual and predicted revenue from the modelling workflow; distinguish the test segment from fitted training predictions.

The existing MMM report records Durbin–Watson 1.76, Ljung–Box p = 0.29, Shapiro–Wilk p = 0.13 and Breusch–Pagan p = 0.23. These tests do not detect strong departures under their respective null hypotheses; with a small sample, this is not proof of independent, normal, constant-variance residuals. It also reports similar CV error across three search seeds.

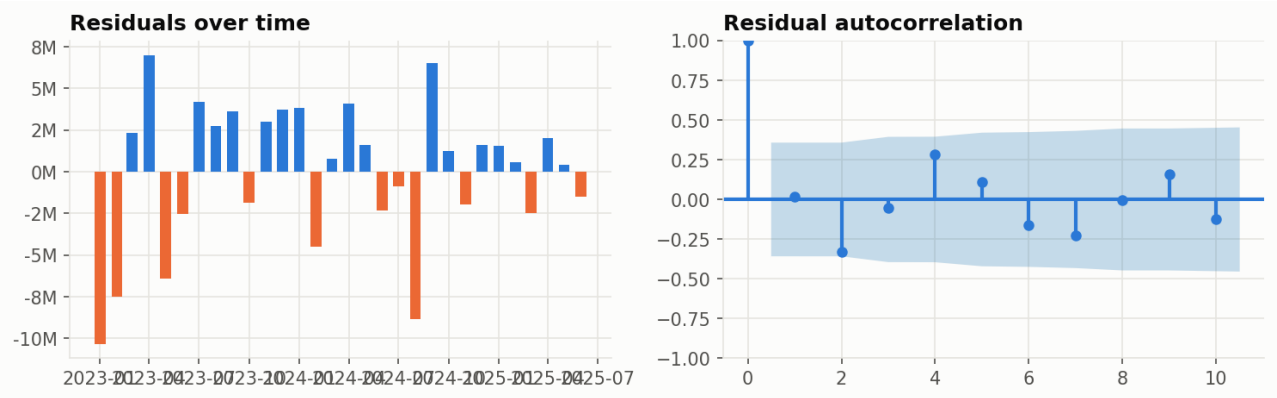


Figure 5. Residual diagnostics.

Uncertainty analysis uses 1,000 moving-block bootstrap samples with three-month blocks. The intervals condition on fixed adstock and saturation settings, so they omit part of total parameter and specification uncertainty. These diagnostics and metrics are consolidated from saved project evidence; this reporting task did not rerun the expensive hyperparameter search.

8. Channel effectiveness and financial interpretation

8.1 Contribution and ROAS definitions

Historical channel contribution is the sum of the fitted channel component over the 36 observed months. Channel ROAS equals this estimated incremental revenue divided by historical channel expenditure. Marginal ROAS is the derivative of the steady-state response curve at the 2025 average monthly spending level.

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Historical channel ROAS = modelled channel contribution / observed channel spend.

Marginal ROAS = change in modelled incremental revenue / small additional change in channel spend.
```

Profit ROI would require incremental contribution profit and relevant costs, for example `(incremental contribution profit − ad spend) / ad spend`. Those inputs are unavailable. Therefore neither ROAS above one nor high predicted revenue establishes profitability.

Channel	Historical spend, LKR m	Modelled contribution, LKR m	Average ROAS	Marginal ROAS	Bootstrap 90% ROAS interval
Outdoor	94.058	575.826	6.12x	1.91	1.91–7.30
YouTube	79.914	332.995	4.17x	1.16	0.10–6.70
Google Search	119.353	486.988	4.08x	1.27	0.15–4.78
TV	156.964	154.091	0.98x	0.48	0.00–2.22
Facebook/Instagram	158.344	113.513	0.72x	0.41	0.00–2.56
Radio	60.108	0.000	0.00x	0.00	0.00–7.34

Estimated media contribution totals approximately LKR 1,663.41 m, or 54.3% of observed revenue. Historical aggregate media ROAS is approximately 2.49x. The remaining revenue is not assigned to media by this specification; it includes the intercept, controls and model residual rather than a verified causal no-advertising outcome.

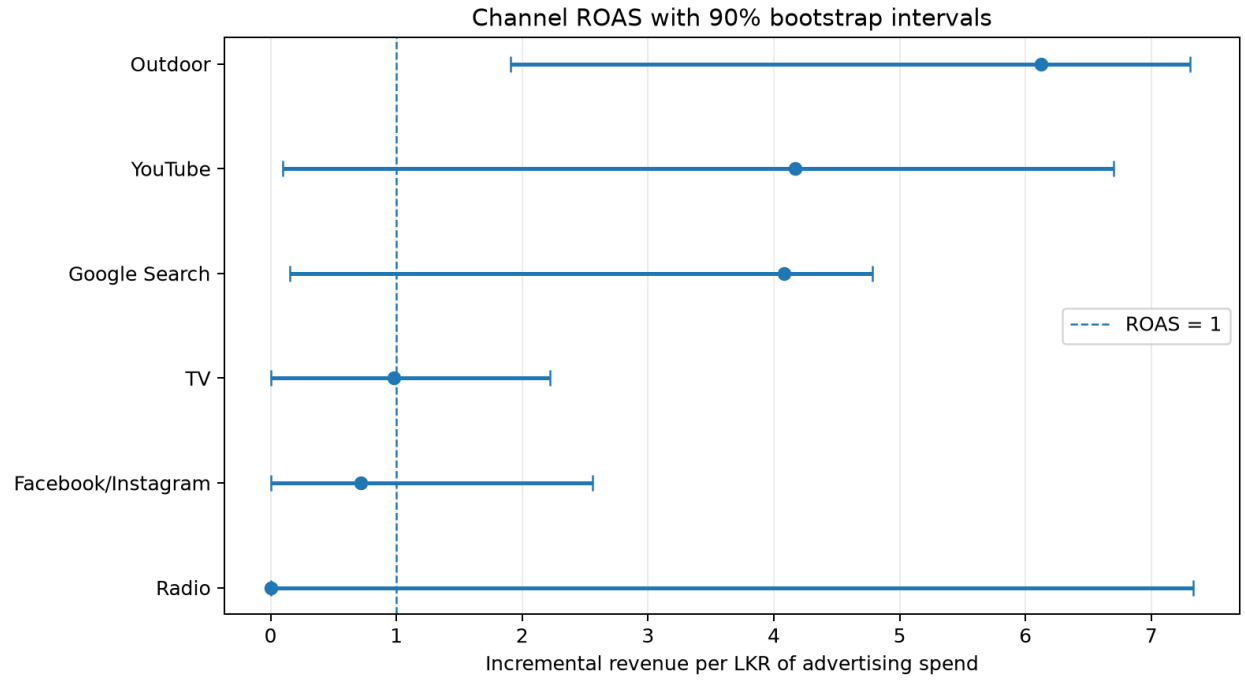


Figure 6. Historical channel ROAS and conditional bootstrap uncertainty.

Outdoor, YouTube and Search receive approximately 44% of historical spend and account for approximately 84% of estimated media contribution. TV and Facebook/Instagram receive approximately 47% of spend and approximately 16% of the modelled effect. Nevertheless, their intervals overlap substantially, and channel rankings should not be treated as exact.

8.2 Carryover, saturation and robust findings

Estimated carryover fractions are approximately 0.38 for Outdoor, 0.30 for TV, 0.22 for Radio, 0.11 for YouTube, 0.08 for Facebook/Instagram and 0.00 for Search. The implied half-lives are below one month. Monthly aggregation and channel-specific parameter bounds limit interpretation of these estimates as precise behavioural durations.

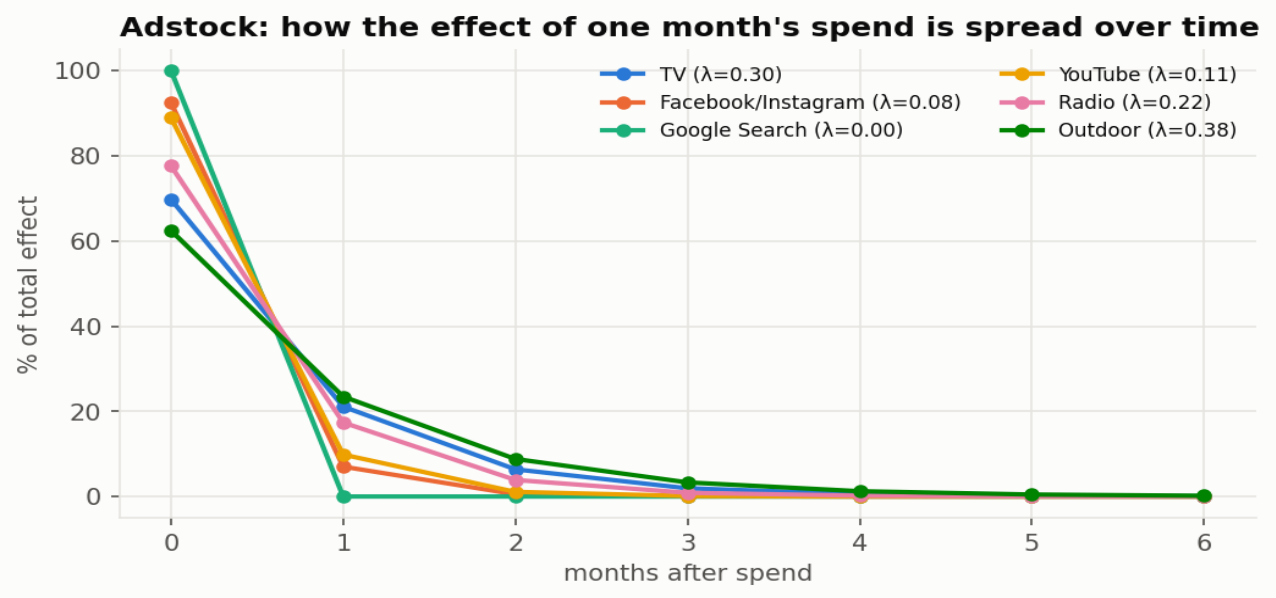


Figure 7. Estimated adstock decay profiles.

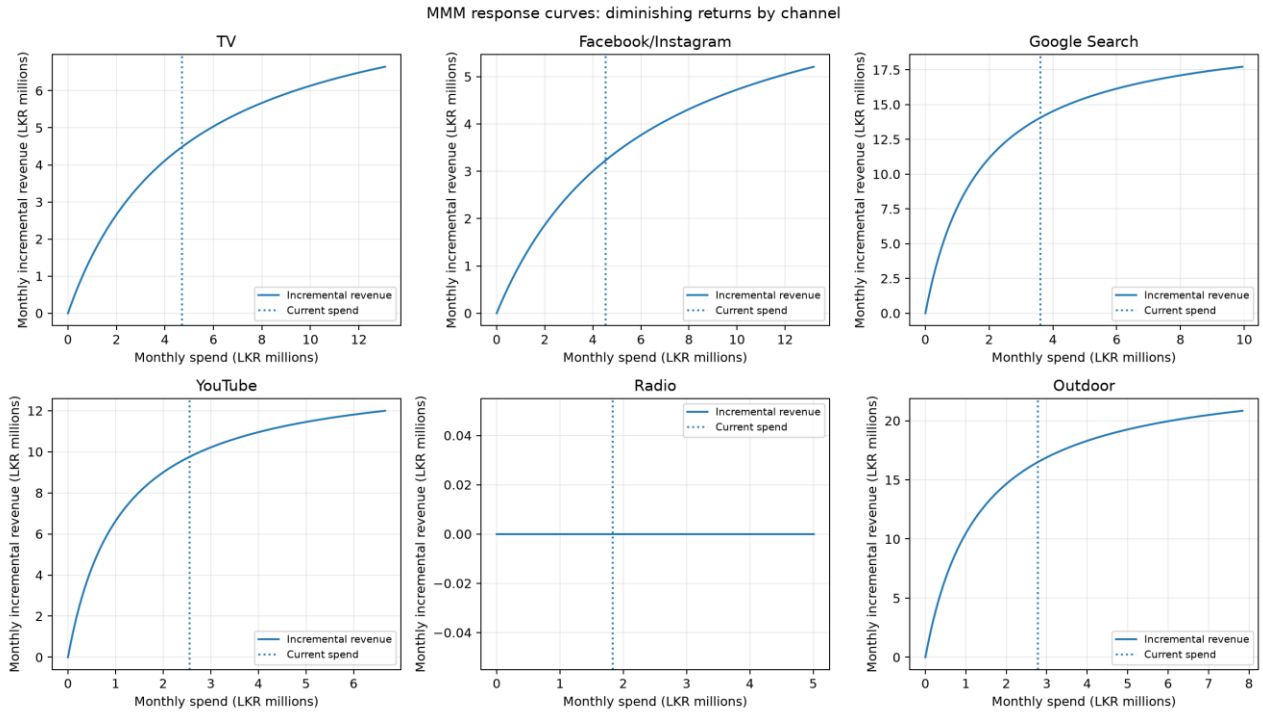


Figure 8. Channel diminishing returns under sustained monthly expenditure.

Search and YouTube rank in the top three across all three attribution methods in the existing comparison. Radio ranks weakest, while Outdoor and Facebook/Instagram vary more by specification. The zero Radio point estimate coexists with a wide bootstrap interval because resampled fits can assign it a positive contribution. This is evidence of uncertainty, not a reason to remove the channel without testing.

9. Budget optimisation methodology and results

9.1 Objective, constraints and solution method

The optimiser loads `models/mmm/final_mmm.json` through `src/mmm_response.py`. It maximises the sum of channel steady-state incremental revenue responses using SciPy SLSQP and an analytical gradient. Variables are scaled to LKR millions for numerical stability.

`Maximise sum(R[c](x[c]))`, subject to `sum(x[c]) = B` and `L[c] ≤ x[c] ≤ U[c]`.

For the selected exponent of one, `R[c](x) = beta[c] × (x / scale[c]) / ((x / scale[c]) + K[c])`. The response is concave for non-negative coefficients; this supports constrained maximisation, but does not remove uncertainty in the estimated response curves.

The baseline budget B is LKR 20,023,825 per month: the sum of the six 2025 average monthly spends. Each channel is restricted to 70–130% of that channel's current average and additionally to its observed historical monthly range. These are illustrative feasibility safeguards, not confirmed contractual or operational limits.

At a constant budget, maximising modelled incremental revenue also maximises modelled portfolio ROAS. External controls are held fixed. The result describes expenditure sustained until carryover settles; it is not a forecast of the first month after changing the allocation.

9.2 Current versus optimised allocation

Channel	Current, LKR m/month	Optimised, LKR m/month	Change	Current marginal ROAS	Optimised marginal ROAS
TV	4.711	4.084	−13.3%	0.48	0.55
Facebook/Instagram	4.541	3.178	−30.0%	0.41	0.54
Google Search	3.601	4.682	+30.0%	1.27	0.88
YouTube	2.553	3.178	+24.5%	1.16	0.84
Radio	1.834	1.284	−30.0%	0.00	0.00
Outdoor	2.782	3.617	+30.0%	1.91	1.32
Total	**20.024**	**20.024**	**0.0%**	—	—

Rounded channel amounts may not add exactly to the displayed total; source CSVs retain full precision. Binding upper and lower constraints mean marginal ROAS need not be equal across all channels at the solution.

Fixed-budget outcome	Current	Optimised
Monthly advertising budget, LKR m	20.024	20.024
Modelled incremental revenue, LKR m/month	48.027	50.143
Portfolio modelled ROAS	2.398x	2.504x

The estimated uplift is LKR 2.116 m/month, or 4.41% of baseline modelled incremental revenue. This percentage is not an uplift in total company sales. Total sales include non-media components that are outside this optimisation objective.

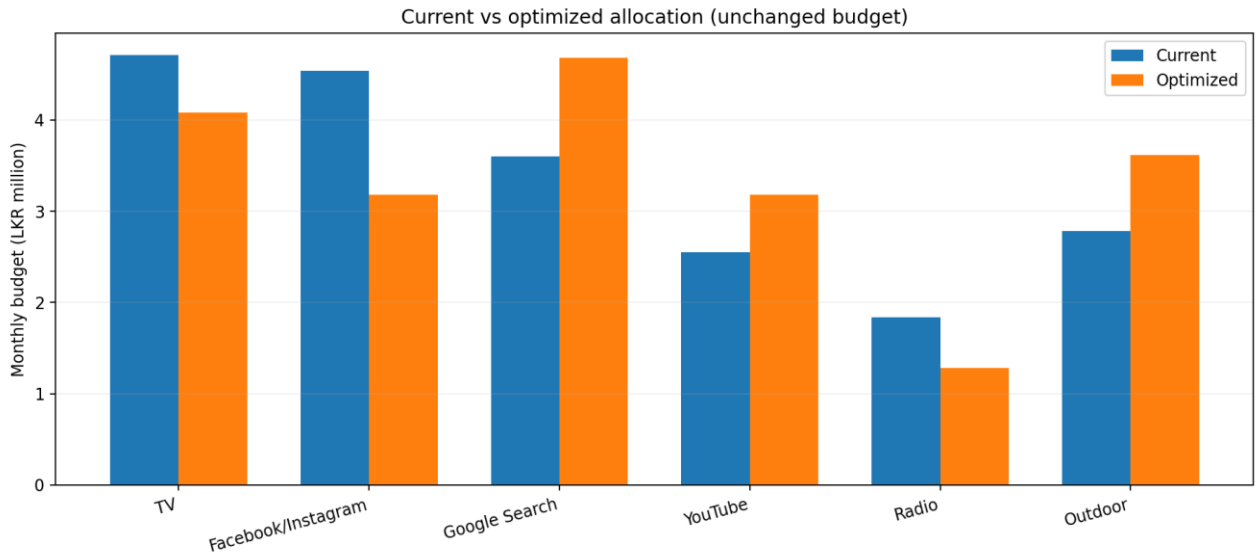


Figure 9. Current and optimised monthly channel expenditure.

9.3 Budget and constraint scenarios

Budget multiplier	Budget, LKR m/month	Optimised incremental revenue, LKR m/month	Portfolio ROAS	Revenue change versus original allocation
0.8x	16.019	45.788	2.858x	−4.66%
0.9x	18.021	48.482	2.690x	+0.95%
1.0x	20.024	50.143	2.504x	+4.41%
1.1x	22.026	51.130	2.321x	+6.46%
1.2x	24.029	51.939	2.162x	+8.15%

Only the 1.0x scenario isolates reallocation at unchanged spend. Other scenarios combine budget changes with reallocation. The 0.9x scenario suggests a potential efficiency opportunity, but its small estimated advantage over the original allocation warrants cautious testing. Revenue rises with larger budgets while ROAS declines, illustrating diminishing returns.

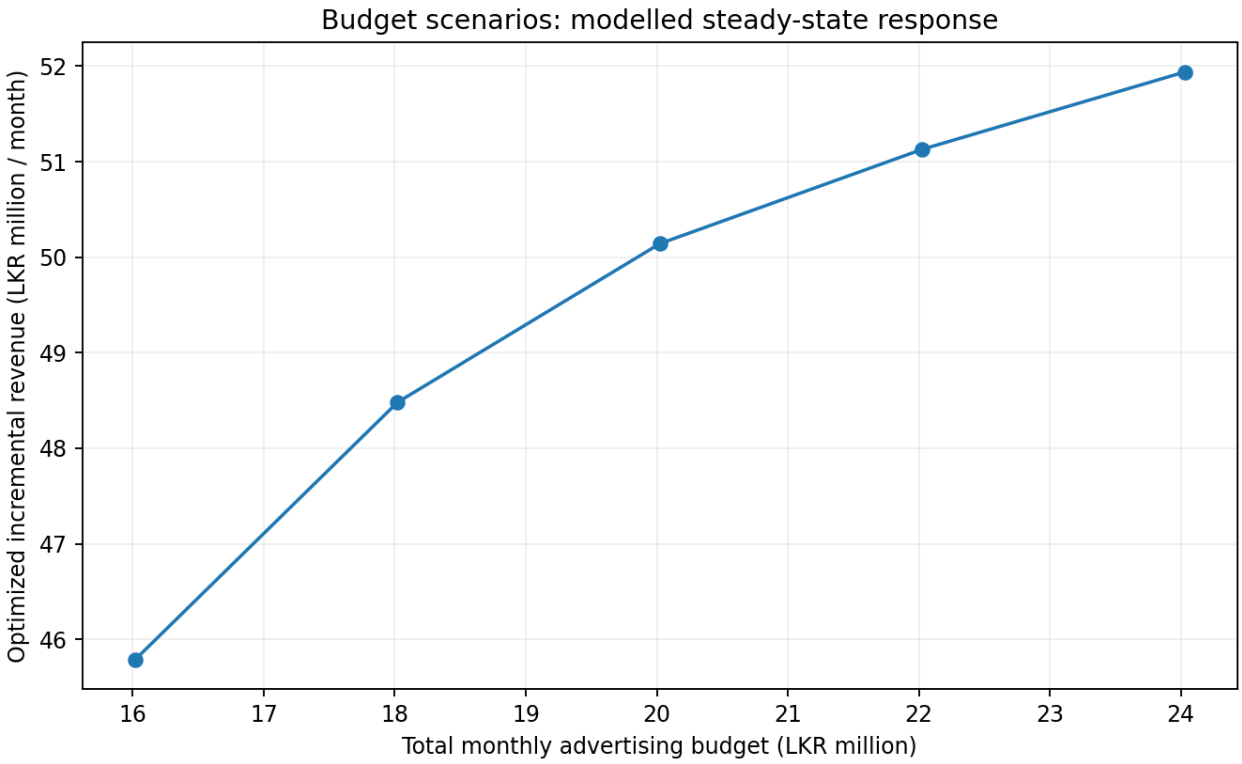


Figure 10. Alternative total-budget scenarios.

Conservative bounds of 85–115% yield a 2.61% modelled fixed-budget uplift; moderate bounds of 70–130% yield 4.41%; flexible bounds of 60–150% yield 4.82%, with historical-range caps also applied. A train-only model sensitivity check estimates approximately 4.88% uplift from its own optimised allocation. These are sensitivity comparisons, not confidence intervals or external validation [9].

10. Dashboard and visual communication

The supplied dashboard artifact is `Dashboard/Dashboard.twbx`, a packaged Tableau workbook containing embedded Hyper extracts. Workbook metadata identifies four dashboard tabs: Executive Summary, Channel Performance, Diminishing Returns and Budget, and Model Evaluation. Four supplied PNG screenshots document the presentation, including a screenshot named Budget Comparison [11].

Requirement deviation: the original brief requests an interactive Power BI dashboard. The completed artifact uses Tableau, consistent with the current README. A Tableau workbook does not itself meet a platform-specific Power BI deliverable. The team should record approval of this substitution or provide an equivalent Power BI file before claiming complete compliance.

10.1 Executive summary

The executive view displays observed total sales, observed total advertising expenditure and blended ROAS, together with revenue trends, budget comparisons and channel recommendations. The screenshot reports sales of LKR 3,063,309,988, spend of LKR 668,740,524 and blended ROAS of 4.581x.

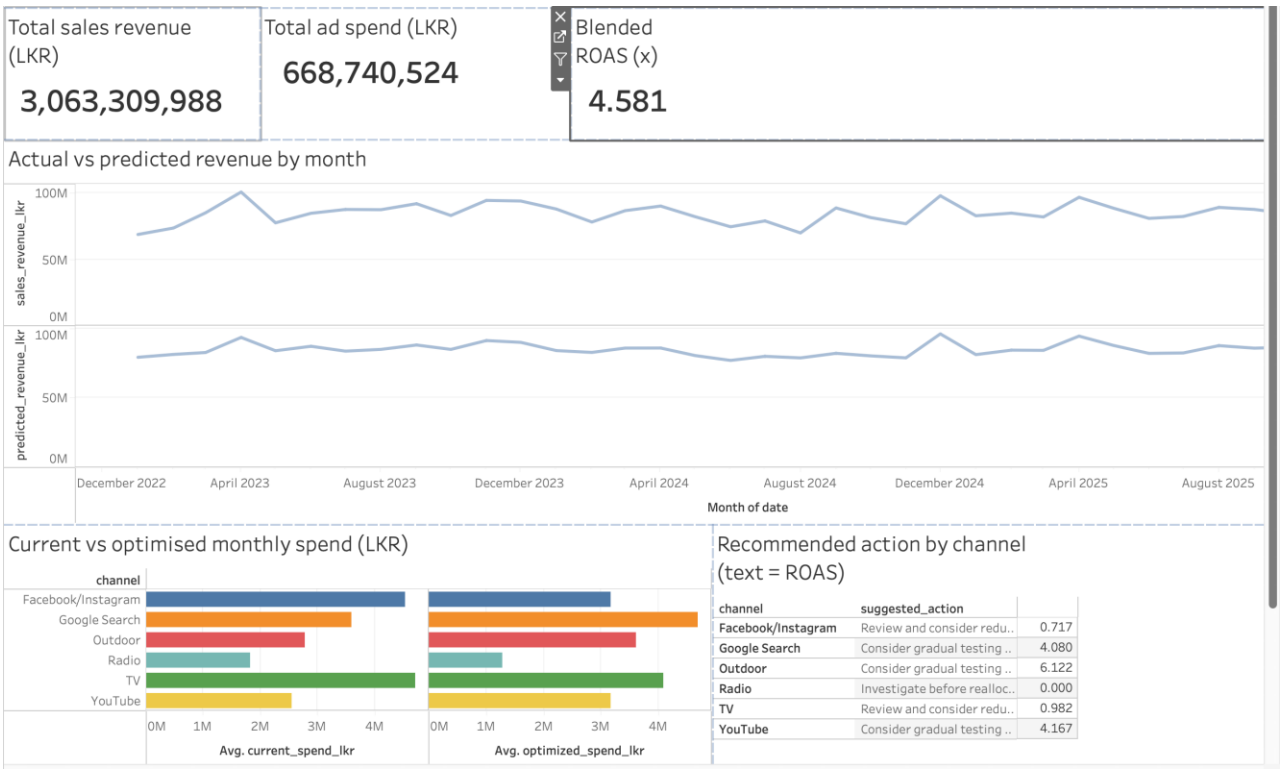


Figure 11. Supplied Tableau Executive Summary.

Blended ROAS is observed total revenue divided by total advertising spend. It includes revenue associated with non-media drivers and must be distinguished from historical model-attributed media ROAS (approximately 2.49x) and the current steady-state optimisation portfolio ROAS (2.398x). Differences between these figures reflect different numerators and reference periods, not inconsistent calculations.

10.2 Channel performance

The channel view supports comparison of contribution, advertising investment and revenue efficiency. Average historical ROAS identifies overall fitted efficiency, while marginal ROAS supports decisions about the next unit of investment. Uncertainty should accompany channel rankings when communicating results to decision makers.

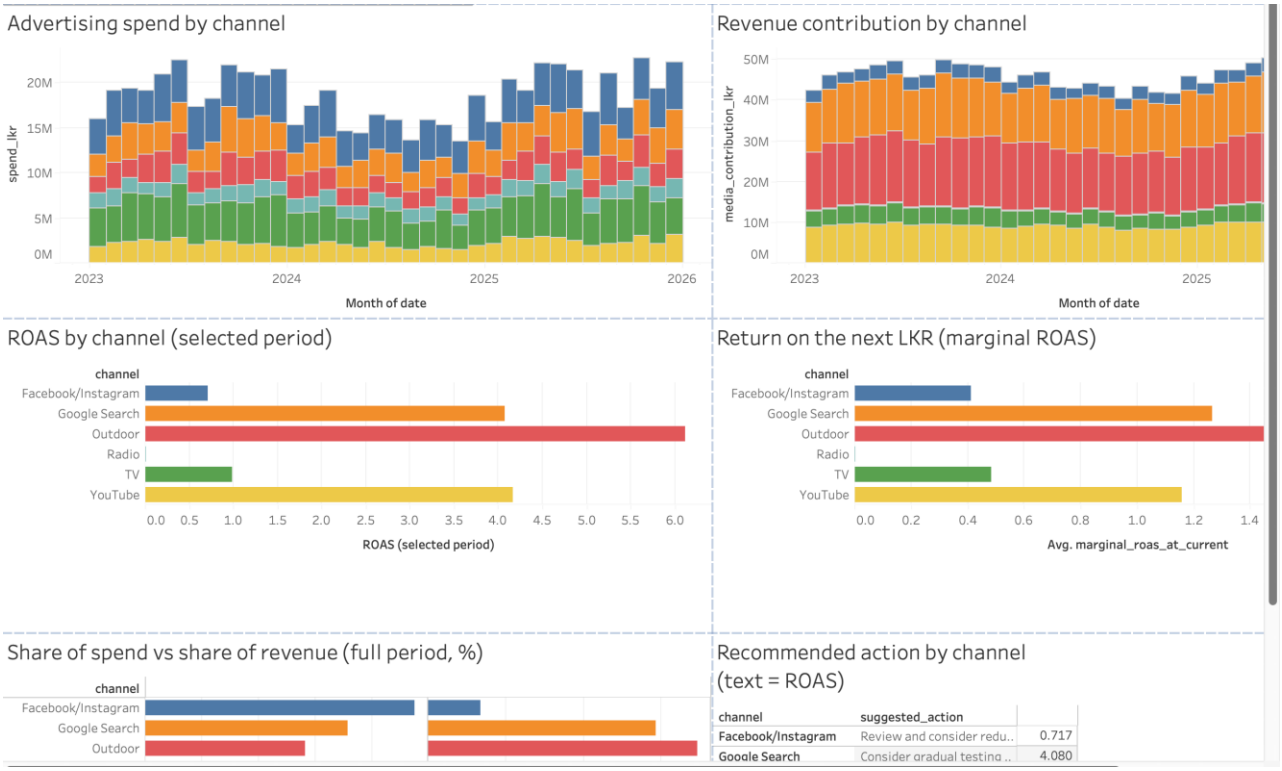


Figure 12. Supplied Tableau Channel Performance.

10.3 Diminishing returns and budgets

Response curves explain why expanding a strong channel does not generate a constant return. Budget comparisons show the current and proposed allocation, and scenario outputs explain the trade-off between total predicted incremental revenue and efficiency.



Figure 13. Supplied Tableau Diminishing Returns and Budget.

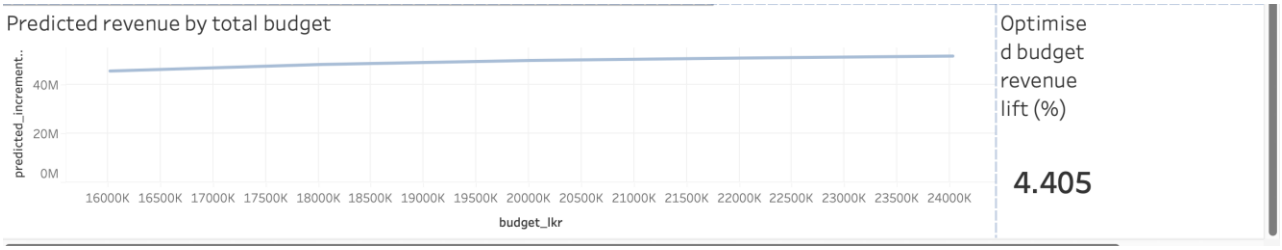


Figure 14. Supplied Budget Comparison.

10.4 Interpretation and verification boundaries

The packaged workbook and screenshots demonstrate an integrated visual deliverable. Interactive date, channel and scenario filters are requested in the brief, but static screenshots do not verify working filter interactions. Filter behaviour and cross-view actions should be demonstrated in Tableau during handover. This report does not claim those interactions were tested. The Model Evaluation tab should present the held-out metrics separately from full-refit historical predictions.

Figures are reproduced as supplied. Some screenshots include the Tableau editing interface and truncated labels; cleaner presentation-mode exports would improve the submission without changing the analysis.

11. Limitations

1. **Small synthetic dataset.** Thirty-six monthly observations limit statistical power and the ability to distinguish correlated channels. Synthetic relationships may not transfer to actual campaigns.

2. **Observational attribution.** Spend can respond to anticipated demand, and unmeasured factors can influence both spend and revenue. Regression attribution does not establish causality.

3. **Limited holdout.** Six test months offer a narrow evaluation window. Overlapping validation folds are also dependent, so ten folds are not ten independent experiments.

4. **Transformation assumptions.** Channel-specific carryover bounds, initial adstock and a fixed Hill exponent influence attribution. Saturation is difficult to identify from narrow spend ranges.

5. **Incomplete uncertainty.** Bootstrap intervals condition on selected transformation parameters and omit some model-selection and specification uncertainty. Wide intervals make exact channel ranks uncertain.

6. **Counterfactual baseline.** No zero-advertising periods are observed. Extrapolating to no media is model-dependent, especially when spend is consistently positive.

7. **Revenue rather than profit.** Missing contribution margins and cost detail prevent profit ROI analysis. Higher ROAS does not ensure a profitable campaign.

8. **Steady-state optimisation.** The allocation assumes sustained spending, fixed external conditions and separable channel curves. It does not explicitly model future shocks, cross-channel interactions, reach limits or campaign execution changes.

9. **Illustrative bounds.** Spending limits require business confirmation. Staying within historical per-channel ranges does not guarantee that the proposed joint allocation has previously been observed.

10. **Delivery and presentation gaps.** Tableau differs from the requested Power BI platform; filter interactions require demonstration; member identities and administrative cover details remain to be entered.

12. Recommendations and conclusion

12.1 Marketing actions

Start with a controlled, conservative reallocation towards Search, YouTube and Outdoor. The 85–115% sensitivity scenario provides a lower-change starting point, with an estimated 2.61% steady-state incremental-revenue uplift under the model. Confirm inventory, campaign quality, contracts and brand requirements before implementing changes.

Use the moderate fixed-budget allocation as a test scenario: approximately LKR 4.084 m for TV, LKR 3.178 m for Facebook/Instagram, LKR 4.682 m for Search, LKR 3.178 m for YouTube, LKR 1.284 m for Radio and LKR 3.617 m for Outdoor per month. These figures are planning estimates under stated assumptions, not immediate spending commitments.

Review TV and Facebook/Instagram campaign quality and strategic objectives alongside their lower fitted returns. Investigate Radio measurement and campaign activity before further cuts; a zero fitted coefficient with a wide interval is insufficient evidence to discontinue a channel. Treat Outdoor's top position cautiously because its rank changes across model specifications.

12.2 Measurement and governance

Action	Suggested owner	Evidence of success
Validate spending floors, caps and campaign capacity	Marketing and finance	Approved feasible allocation and documented assumptions
Run a staged allocation pilot with a credible comparison design	Marketing and analytics	Measured incremental outcomes and uncertainty, allowing for carryover
Collect channel-level margin and incremental cost information	Finance	Profit-based evaluation alongside revenue ROAS
Extend history and record campaign, geographic and creative detail	Data engineering	More independent spend variation and improved data coverage
Refresh evaluation and response curves as new months arrive	MMM team	Stable time-aware accuracy, diagnostics and allocation sensitivity
Resolve dashboard platform and verify filters	Dashboard lead	Accepted platform or Power BI delivery, plus demonstrated interactions

Evaluate pilot results against a comparison group or other credible counterfactual rather than comparing raw before/after sales alone. Account for promotion, seasonality and changing external conditions. Refresh the model after sufficient new observations and reconsider allocations when response estimates or operational constraints change.

12.3 Conclusion

The completed analytical workflow supports a measured shift towards Search, YouTube and Outdoor while retaining caution around attribution uncertainty. M3a balances predictive performance with interpretable non-negative response curves. Under the moderate illustrative constraints, the same LKR 20.024 m monthly budget yields an estimated 4.41% increase in steady-state incremental media revenue. A controlled pilot, verified business constraints and improved financial data are the next steps required to assess whether this opportunity translates into realised value.